

Module 5

Information Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Omni-channel is the integration of different methods consumers use to find and purchase products; these methods include e-commerce websites, mobile apps, e-mail, text messages, social media, phone calls and brick-and-mortar stores.

Answer 1(b)

Hong Kong was described as an e-commerce "wasteland" because it is densely populated and has a wide variety of brick-and-mortar stores in close proximity to population centres. Customers can quickly and easily obtain most of the products and services they need without shopping online.

This situation gave some merchants (especially small and medium-sized companies) little incentive to implement e-commerce and made it difficult for e-commerce businesses to compete against local merchants.

Answer 1(c)(i)

E-commerce businesses often engage in electronic transactions with customers and suppliers with whom they have no direct contact. They may even enter into contracts that are signed electronically. Identity authentication is used to verify the identity of each valid customer before receiving and accepting an order of products and services.

Answer 1(c)(ii)

Payment information may involve a customer's personal information, his/ her credit/ debit card number, security/ authorisation code, and etc. Such information are private and confidential, and they must be secured both in transit and at rest. Digital communication security is used for keeping customers' order and payment information private and safe from unauthorised interception, such as by hackers.

Answer 1(d)

E-business typically involves the use of any two applications as below:

- E-accounting: Performing accounting functions on the Internet. This includes normal accounting tasks, such as processing payroll or accounts receivable data, as well as preparing financial reports or completing income tax returns using online software.
- Customer relationship management ("CRM") system: Software used to collect and store information about existing and potential customers, which is then used to personalise marketing campaigns, improve customer service and manage the sales force.
- Electronic data interchange ("EDI"): Computer-to-computer exchange of business data in structured formats, allowing direct processing of the data by the receiving system.
- Electronic funds transfer ("EFT"): A technology for moving money from one bank account directly to another without the use of paper money or checks.
- Supply chain management ("SCM"): The process of transforming raw materials into a finished product, delivering that product to the consumer and receiving payment.

Answer 2(a)

- (i) Input control
- (ii) Process control
- (iii) Input control
- (iv) Output control
- (v) Process control
- (vi) Process control

Answer 2(b)

Any one of the following is the differences between black hat hackers and white hat hackers:

- Black hat hackers break into networks and computer systems illegally for their own personal gain or to intentionally harm an organisation or individual, while white hat hackers break into networks and computer systems to expose vulnerabilities in an authorised manner; or
- Black hat hackers are interested in stealing valuable information that can be sold on the black market or used to commit other crimes, while white hat hackers are often hired by organisations to help them improve system security.

Answer 2(c)(i)

Intrusion prevention system ("IPS") can be configured to prevent unauthorised and malicious network activities and block the suspicious network traffic via signature-based and anomaly-based detection. Most of the well-known DDoS attacks can be automatically blocked without the need for user intervention, and alarms can also be sent automatically to network administrators for notification on a real-time basis.

Answer 2(c)(ii)

A firewall is used to allow legitimate users to operate within a network with a predefined rule set, while at the same time blocking any unauthorised or even malicious traffic from accessing the network and spreading to other networks. Using firewalls could be able to deny network traffic with known DDoS attack signatures, IP addresses and/or network port numbers.

Answer 3(a)

Any two of the following decision-making techniques could be performed by decision support systems:

- Sensitivity analysis: This type of analysis determines how sensitive an output is to any change in a specific input, so the decision maker can experiment with different inputs to see predicted outcomes.
- What-if analysis: This type of analysis refers to changing assumptions or data in the model to observe the impacts of those changes on the outcome.
- Goal-seeking: In this technique, the decision maker has a specific outcome in mind and needs to figure whether it is feasible to achieve that desired outcome and, if it is feasible, how to achieve that outcome.
- Risk analysis: This type of analysis uses the decision support system to evaluate the risk of alternative strategies or actions.

Answer 3(b)

Managers may use their information systems to retrieve data and information from customer relationship management ("CRM") system, transaction processing system ("TPS"), management information system ("MIS"), and/ or enterprise resource planning ("ERP") system, to facilitate any two performance management activities in sales and marketing as follows:

- matching actual monthly/ quarterly/ annual sales revenue with target sales amount;
- monitoring monthly/ quarterly/ annual sales growth;
- evaluating performance per each salesperson or sales team; and/ or
- performing profitability analysis on customers and products.

Answer 3(c)

It is necessary to establish a code of ethical conduct for using information systems because:

- the code communicates to all parties in an organisation about the principles of ethical information systems use, which can be used as an educational mechanism to point employees in the right direction; and
- it also defines the approach adopted by an organisation to ethics, which provides a framework for preventing, detecting and responding to unethical choices made by members of the organisation.

Answer 3(d)

Auditors should be familiar with any two of the following standards related to IT security:

- Hong Kong Standard on Auditing 315 (Revised) *Identifying and Assessing the Risks of Material Misstatement through Understanding the Entity and Its Environment*, which defines objectives, guidelines and processes for auditing information systems. The auditor is expected to gain a certain understanding of the information system, including related processes, as they relate to financial reporting;
- ISO 27001 *Information technology – Security techniques – Information security management systems – Requirements*, which outlines the high-level requirements for an effective information security management system; and
- ISACA's Standards for IS Audit and Assurance which define mandatory requirements for IS auditing and reporting, and inform IS audit and assurance professionals of the minimum level of acceptable performance required to meet the professional responsibilities.

Answer 4(a)

Being a data owner, he/ she has any two of the following responsibilities:

- ensuring data accuracy, as data are usually retrieved and used for generating reports;
- deciding who is/ are legitimate user(s), as the permission of accessing data should be granted to authorised staff only;
- ensuring the data are stored properly, as it is usually shared to and used by different departments and systems;
- establishing and providing accountability, as it enables the organisation to trace and track activities performed in a database or a data file;
- maintaining data security to ensure data confidentiality and privacy, as sensitive and/ or personal data may be stored among databases and data files, and they must be protected from unauthorised access based on regulatory and contractual requirements; and
- ensuring only relevant data are entered and stored into data files so as to prevent excessive amounts of data are being stored and used for further analysis.

Answer 4(b)

A storage area network ("SAN") is a combination of software and hardware configured to run on a separate network of devices dedicated to have a very large volume of data storage, and it is usually used by data centers or big organisations with dozens or hundreds of servers. However, a network attached storage ("NAS") is a server/device connected to a computer network for general data storage, and it is usually used by small and medium-sized organisations or individuals.

Answer 4(c)

Any two of the following benefits would be obtained by using an extranet in supply chain management:

- eliminating some operating and administrative costs through system automation;
- reducing manpower and its associated costs;
- enhancing data accuracy through paperless operations;
- shortening processing time and improving processing efficiency; and
- providing convenience to connect business partners around the world.

Answer 4(d)

Any two of the following prediction outcomes could be generated from Big Data:

- the likelihood that an e-mail message is spam;
- the likelihood that a typo mistake occurs in a message;
- the likelihood that a person is jaywalking; or
- the likelihood that a fraudulent credit card transaction exists.

Answer 5(a)

From design to implementation and even into the maintenance phase, an accountant could serve as a subject expert. He/ she may provide information and insight into how data enters the system, how that data is manipulated to produce the desired outputs and how data and information are used. He/ she may also know the controls that must be in place to ensure compliance and to protect data and system integrity.

Answer 5(b)

Any two of the following techniques may be used to address resistance to a new accounting system:

- revising management policies, such as updating the formal standard operating procedures to describe how work is to be performed with the new system;
- analysing the costs and benefits for each set of potential adopters; this may help pinpoint those who will likely support the change and those who will not;
- motivating adoption, such as rewarding early adopters with a cash bonus or time off, and motivating late adopters by showing them the success of early adoption;
- enabling staff to adopt through training, such as providing the skills needed to adopt and helping users to accomplish their jobs;
- creating a culture of curiosity, learning and experimentation that promotes early adoption of new technologies and methods;
- involving end users in the system development life cycle. If users serve on the IT steering committee and serve as testers, then they are more likely to embrace the change and encourage others to do so; or
- obtaining executive or senior management sponsorship. If leadership conveys the belief that the new information system is essential for the organisation's success, then everyone will be more motivated to adopt it.

Answer 5(c)

Change management is a process of managing changes to a system to make sure changes are authorised and ensure that the information system continues to meet the organisation's business needs.

The change management process itself can be considered a quality assurance process because it:

- ensures the changes are necessary and do not create any potential or possible login/data error(s);
- maintains separate environments for development, testing and production, and reduces the risk of errors being introduced into the system used for daily operations; or
- enforces segregation of duties between programmers and testers, and ensures the changes are properly made and tested.

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